

- Summarize the pathogenesis of **Kawasaki disease, rheumatic fever/disease, cardiomyopathies** and **syncope** in the pediatric population
- Describe the clinical presentation and management of these disease states
- Construct a working differential diagnosis for these diseases
- Be able to discuss the treatment for these disease states

SESSION OBJECTIVES

- SYNCOPE = involuntary loss of posture & consciousness
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- Many causes:
 - Vasovagal-interruption of normal central vasomotor tone leading to arteriolar vasodilation, hypotension, and decreased cerebral perfusion pressure. “Common faint” may be more common after fasting, associated w/anxiety, fright, surprise.
 - Vagovagal-when vagal stimulation causes extreme bradycardia, i.e., suctioning a newborn’s posterior pharynx
 - Cardiac-usu. associated w/heart lesions that cause obstruction to L-sided blood flow, ex: aortic stenosis
 - Arrhythmogenic-arrhythmias ex: long QT syndrome
 - Orthostatic- failure of vascular compensatory responses to postural changes leading to a decrease in BP and cerebral perfusion, ex: dehydration

LONG QT SYNDROME- Arrhythmogenic cause of syncope

- Myocardial repolarization disorder
- May progress to ventricular arrhythmias(torsades de pointes)- then syncope
- May lead to sudden death
- QT interval usu. $>.46$ sec
- May be congenital or acquired, F>M
- Congenital: Romano-Ward Syndrome(nl hearing, autosomal dominant), Jervell and Lange-Nielsen Syndrome(deafness, autosomal recessive), Anderson-Tawil Syndrome(autosomal dominant), Timothy Syndrome(sporadic)

Ion channel disturbances, mutations in the genes responsible for Ca^{+} , K^{+} , and Na^{+} are implicated in congenital LQTS

- Acquired: drugs, electrolyte disturbances, underlying disease like hypothyroidism
- May have history or FHx of congenital deafness, +/-or sudden death

Syncope in Children (work-up)

- Most are vasovagal or noncardiogenic, last less than 1 minute
- Use history to your advantage:
 - Incontinence: prob seizure (non-cardiac)
 - Prior cardiac surgery: prob arrhythmia
 - Family history of sudden cardiac death/congenital deafness: prob arrhythmia
 - Exercise-induced: prob hypertrophic obstructive cardiomyopathy or arrhythmia
 - Longer duration: prob structural cardiac or arrhythmia
 - Witnessed cardiac arrest: probably not neurologic

HPI-case #2

A 7 yo female presents for her 1st visit to the pediatrician's office since arriving in the US.

3 weeks ago, she presented to Border Control in Texas because of a 2-day history of fever with a Tmax of 104°F oral (40°C) and a sore throat. Symptomatic treatment recommended.

Physical exam in Texas revealed:

- Enlarged, slightly tender b/l anterior cervical lymph nodes
- Oropharyngeal injection with soft palate petechiae
- Rapid strep. test neg, no throat culture sent
- CVR: increased rate, nl rhythm. Lungs CTA no RRW
- Skin: no rash

Today, she presents with a 1-week history of intermittent/fleeting and random knee and wrist swelling accompanied by pain and redness. Decreased activity, tired, and developed a flat pink arm rash with a pale center 2 days ago.

Rheumatic Fever (Treatment)

- Best treatment = prevention
 - Antibiotics used for strep pharyngitis is for **ARF prevention**, not treatment for the pharyngitis
- Once diagnosed with ARF, long-term penicillin prophylaxis
 - Oral twice daily
 - IM once monthly
- Treatment of ARF = eradication of strep using penicillin, plus NSAIDs +/- steroids
- Treatment of CHF / valvular disease as necessary
- NSAIDs may mask migratory polyarthrititis, best to use APAP prior to diagnosis

- a. Kawasaki disease
- b. Acute rheumatic fever
- c. Truncus arteriosus
- d. Hypoplastic left heart syndrome
- e. Dilated cardiomyopathy

DIAGNOSIS?

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DIAGNOSIS

Myocardial Diseases: based on etiology

A. Infectious-Viral, nonviral, immunologic(Kawasaki syndrome, acute rheumatic fever)

B. Metabolic-

1.Electrolyte-
hypoglycemia,magnesemia,kalemia and hyperkalemia

2.Endocrine-
hypothyroid,pheochromocytoma, infant of diabetic mother

3.Drugs-
etoh,cocaine,radiation,chloroquine,adriamycin

4. Vitamin/trace metal deficiencies-
thiamine, selenium, carnitine, taurine

C. Infiltrative-Glycogen storage/glycolipid diseases, hemochromatosis, amyloidosis, sarcoidosis, neoplasms, mucopolysaccharidoses

D. Ischemic- rare, congenital anomalies of coronary arteries

Cardiomyopathy: Disease of the myocardium: usu have a large heart(more muscle or larger cavity)

Primary Myopathy Classification:

- DILATED-most common, may be idiopathic, infectious (Coxsackie B virus or echovirus), postinfectious, familial, secondary to systemic disease or cardiotoxic drugs
- HYPERTROPHIC- familial 50% of patients
- RESTRICTIVE-rare in pediatrics, idiopathic, secondary to systemic disease

Dilated Cardiomyopathy-most common

- Myocyte atrophy increases left ventricular(LV) cavity, decreases contractility
 - Low ejection fraction(EF)- Large LV with decreased pump ability
 - Pumping defect = systolic dysfunction
- Presents with CHF
 - Decreased cardiac output
 - May present with LBBB
- Identify cause, if possible, to help treatment-but often idiopathic
 - Reversible (etoh, infection, pregnancy, malnutrition)
 - Irreversible (doxorubicin, genetic)
- Treatment:
 - Treat causative agent (if possible) and symptoms(CHF, heart block)
 - Heart transplant
- Prognosis: 50% mortality within 5 years

Hypertrophic Cardiomyopathy(HCM)

- Myocyte hypertrophy decreases LV cavity size
 - High EF (LV cannot fill, but pumps it all out)
 - Filling defect = Diastolic dysfunction
- Asymptomatic until sudden cardiac death
 - Increased HR (running / sports) can cause arrhythmia in pathologic myocardium
- Treatment: Symptomatic and
 - Myomectomy-resection of subaortic muscle may help symptoms
 - Heart transplant
- Prognosis: sudden death 5% pts per year

Restrictive Cardiomyopathy-Rare

- Diastolic dysfunction(extremely poor compliance of the ventricular walls)
- Ventricular volume is normal to mildly decreased
- Wall thickness is normal to minimally increased
- Typically, atrial enlargement
- Treatment: control symptoms and maintain BP control without decreasing cardiac output, cardiac transplant
- Prognosis: poor, 5-year survival <30%

Case #4 Physical Exam

- VS: 104.4F, HR 160, RR 30, pulse ox 96%
- Nontoxic appearing, crying but consolable in mom's arms, WH, hot to touch, WD
- Eyes: conjunctiva- injected b/l, no discharge, PERRLA, EOMI
- Mouth: no oral ulcers, + injection of the oropharynx , tongue with red, prominent papillae, lips-cracked and erythematous
- Ears: TM's pink, flat, positive light reflex
- Neck: supple, 2 x 2cm right tonsillar lymph node, mobile, nontender
- CVR: increased HR, no murmur, lungs: CTA no RRW
- Ext: hands/feet: nonpitting edema
- Rapid strep. test neg. in office, throat culture is pending

- a. Scarlet fever
- b. Stevens-Johnson syndrome
- c. Juvenile idiopathic arthritis
- d. Kawasaki disease
- e. Acute rheumatic fever

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DIAGNOSIS

Kawasaki Disease KD (Epidemiology)

- Generalized vasculitis of unknown etiology; ?mimicry of an infectious agent
- Has endemic & epidemic patterns: most common in winter/spring
- More common in children (80% < 8 years), more common in Asian ancestry
- A leading cause of acquired heart disease in children

Kawasaki Disease (Diagnosis)

- Fever >5 days PLUS 4 of the following:
 1. Changes in the extremities: Acute : erythema/edema of the hands/feet
Subacute: desquamation hands/feet
 2. Changes in the oral mucosa (include red/cracked lips, injected oral mucosa and pharynx, strawberry tongue)
 3. Polymorphous exanthema
 4. Bilateral conjunctival injection w/o pain or exudate
 5. Unilateral cervical lymphadenopathy
- Lab findings may include elevated WBC's, anemia, normal plt count early and thrombocytosis late
- Coronary aneurysms are a major concern in untreated disease

Strawberry Tongue

- KD causes white exudate on the tongue. Papillae enlarge due to erythema & inflammation
- Appearance changes as white exudate disappears along with desquamation of the tongue surface (with a decrease in sense of taste)
- Causes of strawberry tongue include (from most to least common):
 - Scarlet fever (group A beta-hemolytic strep with enterotoxin)
 - Kawasaki disease
 - Toxic-like shock syndrome



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Kawasaki Disease (Symptoms)

- Disease has 3 phases:
 - * 1. Acute Febrile –usu. 1- 2 wks. of illness(unremitting fever lasts usu 1-2 weeks but may persist up to a month), necrotizing arteritis, saccular aneurysms may form
 - ** 2. Sub-acute- usu lasts 3-4 weeks(subacute/chronic vasculitis which may last years), desquamation, thrombocytosis
 - ** 3. Chronic or “convalescent”-lasts up to 2 mos. Usu resolves 6-8wks after onset. Vessels of vasculitis develop smooth muscle myofibroblasts leading to stenosis
- Most symptoms(*) are apparent in acute phase, more serious sequelae(**) occur in sub-acute or chronic phase
- Thus, early diagnosis and treatment is of the utmost importance

Kawasaki Disease-untreated (Coronary artery disease stages)

- Stage 1: (first 2 wks.) acute **vasculitis** of coronary arteries.
- Stage 2: (2-4 wks. after onset) vasculitis persists, may lead to a weakening of the wall and **aneurysms** may occur. Flow decreases with the increasing diameter. Stagnant blood in aneurysm may lead to thrombosis. Thrombus in coronary arteries may lead to obstruction of coronary blood flow.
- Stage 3: (4-8wks after onset) artery inflammation subsides but aneurysms may persist.
- Stage 4: (>8wks after onset) scar formation and **calcification** of the arteries with stenosis and recanalization of the lumen, myocardial fibrosis.
- 20% of untreated children will develop coronary artery disease.

Kawasaki Disease (Treatment)

- Initial therapy: Reduce inflammation and prevent coronary aneurysms!
- Chronic therapy: Prevent coronary artery thrombosis
- IVIG (intravenous immunoglobulin): first-line tx'ment to decrease immune response-decreases risk of coronary artery aneurysm if used in the first 10 days of symptoms. Approx. 85% success rate.
- ACUTE(anti-inflammatory): Aspirin 80-100mg/kg/d divided QID: reduces systemic symptoms and decreases risk of aneurysm formation
- CHRONIC(antithrombotic): Aspirin 3-5mg/kg qD when afebrile and discontinued 6-8wks after the onset of illness UNLESS coronary aneurysms are present(continue low dose aspirin indefinitely)
- Second dose IVIG or steroids or TNF-alpha antibody are used in more resistant cases.

- **Syncope**: Must identify cause to prevent subsequent episodes
- **Cardiomyopathy**: Intrinsic disease of the heart muscle
- **ARF**: Strep pharyngitis sequela; JONES criteria for diagnosis; damages valves; requires long-term prophylaxis to prevent recurrence
- **Kawasaki disease**: Vasculitis; clinical diagnosis; treatment focuses on coronary aneurysm prevention and symptomatic treatment